EDS223 - Homework Assignment 3

Identifying Ideal Areas for Oyster and Shrimp Development

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This assignment looks at ocean depth and sea surface temperature to calculate the most appropriate zones to cultivate oysters and shrimp. The greater region of study is the Pacifc Coast of the United States, reaching from the Canadian border in the North to the Mexican border in the South. For the shrimp, the most suitable area for cultivation is off the coast

of Oregon. For oysters, the most suitable area for cultivation is off the Central Coast of California.

1 Load libraries

```
library(sf)          # vector handling
```

```
Linking to GEOS 3.13.0, GDAL 3.8.5, PROJ 9.5.1; sf_use_s2() is TRUE
```

```
library(terra)       # raster handling
```

```
terra 1.9.50
```

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
```

```
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x tidyr::extract() masks terra::extract()
x dplyr::filter()  masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to beco
```

```
library(tmap)        # map making
```

```
Registered S3 method overwritten by 'stars':
  method                  from
  st_interpolate_aw.stars sf
```

```
library(elevatr)     # elevation data download
```

```
elevatr v0.99.0 NOTE: Version 0.99.0 of 'elevatr' uses 'sf' and 'terra'.  Use
of the 'sp', 'raster', and underlying 'rgdal' packages by 'elevatr' is being
deprecated; however, get_elev_raster continues to return a RasterLayer.  This
will be dropped in future versions, so please plan accordingly.
```

```
library(tigris)      # US Census boundary data
```

```
To enable caching of data, set `options(tigris_use_cache = TRUE)`
in your R script or .Rprofile.

Attaching package: 'tigris'

The following object is masked from 'package:terra':

    blocks
```

```
library(viridisLite)
library(kableExtra) # table formatting
```

```
Attaching package: 'kableExtra'

The following object is masked from 'package:dplyr':

    group_rows
```

```
library(spData) # spatial data
library(spDataLarge) # spatial data
library(geodata) # spatial data
library(stars)
```

```
Loading required package: abind
```

2 Load the data

```r
average_sst_2008 <- rast('data/data/average_annual_sst_2008.tif')
average_sst_2009 <- rast('data/data/average_annual_sst_2009.tif')
average_sst_2010 <- rast('data/data/average_annual_sst_2010.tif')
average_sst_2011 <- rast('data/data/average_annual_sst_2011.tif')
average_sst_2012 <- rast('data/data/average_annual_sst_2012.tif')
depth <- rast('data/data/depth.tif')
regions <- st_read('data/data/wc_regions_clean.shp')
```

```
Reading layer `wc_regions_clean' from data source
  `/Users/elyseowen/Desktop/School/UCSB/MEDS/eds223/EDS223-HW3/data/data/wc_regions_clean.shp
  using driver `ESRI Shapefile'
Simple feature collection with 5 features and 5 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -129.1635 ymin: 30.542 xmax: -117.097 ymax: 49.00031
Geodetic CRS:  WGS 84
```

3 Stack the average SST rasters together, calculate the mean, convert to from Kelvin to Celcius

```r
average_sst_stack <- c(average_sst_2008, average_sst_2009, average_sst_2010, average_sst_201
mean_sst <- mean(average_sst_stack, na.rm = TRUE)
average_sst_c <- mean_sst - 273.15
```

4 Crop depth raster to the extent of the SST raster, resample to the depth raster. C

```r
sst <- ext(-131.9848, -114.9867, 29.99305, 49.98842)
depth_cropped <- crop(depth, sst)
depth_resampl <- terra::resample(depth_cropped, y = average_sst_c, method = "bilinear")
average_sst_c <- project(average_sst_c, crs(depth_resampl))
```

5 Check that the CRSs match

```
cat("\n=== Same CRS? ===\n")
```

```
=== Same CRS? ===
```

```
cat("west cost regions vs depth: ", same.crs(regions, depth_resampl), "\n")
```

```
west cost regions vs depth:  TRUE
```

```
cat("west cost regions vs SST stack: ", same.crs(regions, average_sst_c), "\n")
```

```
west cost regions vs SST stack:  TRUE
```

```
cat("depth vs SST stack: ", same.crs(depth_resampl, average_sst_c), "\n")
```

```
depth vs SST stack:  TRUE
```

6 Build function to classify ideal regions for aquaculture

```
Species_Regions <- function(minimum_sst,maximum_sst,minimum_depth,maximum_depth,
species_name) {
    rcl_temp <- matrix(c(-Inf, minimum_sst, 0, # not suitable
                    minimum_sst, maximum_sst, 1,  # suitable
                    maximum_sst, Inf, 0), # not suitable
                  ncol = 3, byrow = TRUE)

    # Apply the matrix to reclassify the raster, making all cells 0 or 1 or 2
    temp_rcl <- terra::classify(average_sst_c, rcl = rcl_temp)

    # Define a reclassification matrix for temperature
    rcl_depth <- matrix(c(-Inf, minimum_depth, 0, # not suitable
                    minimum_depth, maximum_depth, 1,  # suitable
                    maximum_depth, Inf, 0), # not suitable
```

```
                ncol = 3, byrow = TRUE)

    # Apply the matrix to reclassify the raster, making all cells 0 or 1 or 2
    depth_rcl <- terra::classify(depth_resampl, rcl = rcl_depth)

    combined <- c(temp_rcl, depth_rcl)

    suitable_zones <- lapp(combined, fun = function(SST,depth) SST * depth)

    regions_rasterise <- rasterize(regions, suitable_zones, field = 'rgn_id')

    suitable_cellsize <- cellSize(suitable_zones, unit = "km", mask = TRUE)

    suitable_zones_mask <- mask(suitable_cellsize, suitable_zones, maskvalues = 0)

    zone <- zonal(suitable_zones_mask, regions_rasterise, fun = "sum", na.rm = TRUE)

    zone_tibble <- as_tibble(zone)

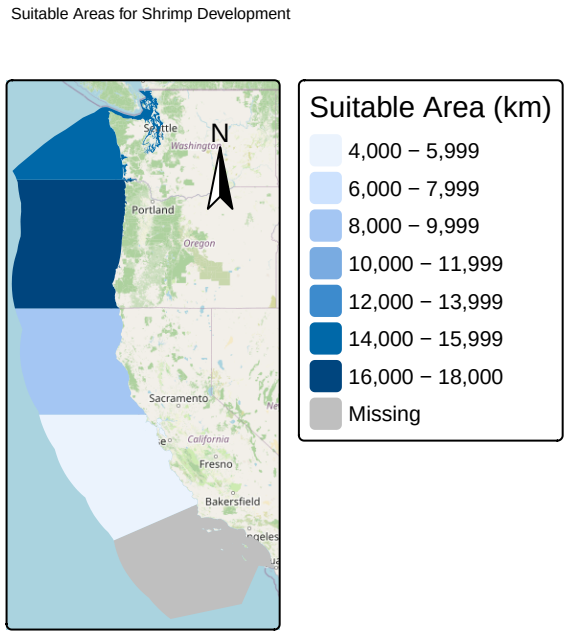
    zone_map <- left_join(regions, zone_tibble, by= 'rgn_id')

    tm_shape(zone_map)+
  tm_basemap('OpenStreetMap')+
  tm_fill(
    fill = 'area',
    fill.scale = tm_scale(),
    fill.legend = tm_legend(title = 'Suitable Area (km)',
    position = tm_pos_out('right')))+
    tm_title(species_name)+
    tm_compass(position = c('top', 'right'))
}
```

7 Run function on shrimp

```
Species_Regions(minimum_sst = 6, maximum_sst = 12, minimum_depth = -457, maximum_depth = -36
```

```
[plot mode] fit legend/component: Some legend items or map compoments do not
fit well, and are therefore rescaled.
i Set the tmap option `component.autoscale = FALSE` to disable rescaling.
```

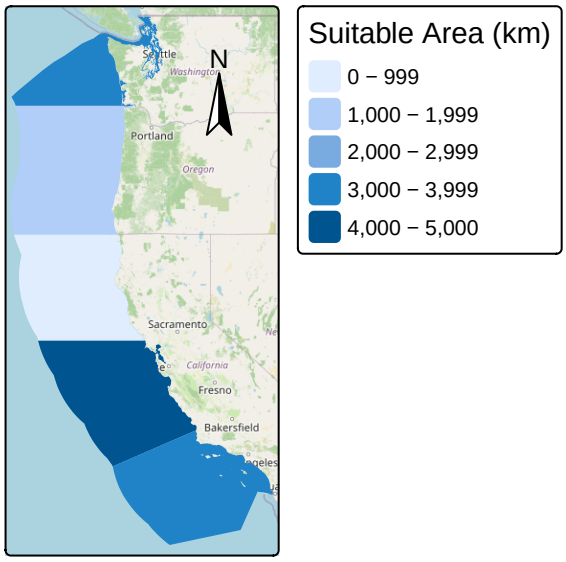


8 Run function on oysters

```
Species_Regions(minimum_sst = 11, maximum_sst = 30, minimum_depth = -70, maximum_depth = 0, s
```

```
[plot mode] fit legend/component: Some legend items or map compoments do not
fit well, and are therefore rescaled.
i Set the tmap option `component.autoscale = FALSE` to disable rescaling.
```

Suitable Areas for Oyster Development



9 Components of building the function

9.1 Create matrix of ideal temperature, depth for oyster development. Output produces suitable zones for oyster development.

```
rcl_temp <- matrix(c(-Inf, 11, 0,
                     11, 30, 1,
                     30, Inf, 0 ),
                     ncol = 3, byrow = TRUE
                     )
sst_rcl <- terra::classify(average_sst_c, rcl = rcl_temp)

rcl_depth <- matrix(c(-Inf, -70, 0,
                     -70, 0, 1,
                     0, Inf, 0),
                     ncol = 3, byrow = TRUE
                     )
depth_rcl <- terra::classify(depth_resampl, rcl = rcl_depth)

combined <- c(sst_rcl, depth_rcl)
```

```
suitable_zones <- lapp(combined, fun = function(SST,depth) SST * depth)
```

9.1.1 Rasterise the regions, suitable_zones on 'rgn_id'. Calculate ideal cells with masking, summarise and put into a tibble.

9.1.2 Join the summarised zones with regions, create the final map.

```
sf::st_crs(regions) <- terra::crs(suitable_zones)
regions_rasterise <- rasterize(regions, suitable_zones, field = 'rgn_id')

suitable_cellsize <- cellSize(suitable_zones, unit ='km', mask = TRUE)

suitable_zones_mask <- mask(suitable_cellsize, suitable_zones, maskvalues = 0)

zone <- zonal(suitable_zones_mask, regions_rasterise, fun = 'sum', na.rm=TRUE)

zone_tibble <- as_tibble(zone)

zone_map <- left_join(regions, zone_tibble, by = 'rgn_id')

tm_shape(zone_map)+
  tm_basemap('OpenStreetMap')+
  tm_fill(
    fill = 'area',
    fill.scale = tm_scale(),
    fill.legend = tm_legend(title = 'Suitable Area (km)',
    position = tm_pos_out('right')))+
  tm_title('Suitable Areas for Oyster Development')+
  tm_compass(position = c('top', 'right'))
```

```
[plot mode] fit legend/component: Some legend items or map compoments do not
fit well, and are therefore rescaled.
i Set the tmap option `component.autoscale = FALSE` to disable rescaling.
```

Suitable Areas for Oyster Development

